

DDB worksheet

Exercise 1

Tuto 3 — Distributed DB



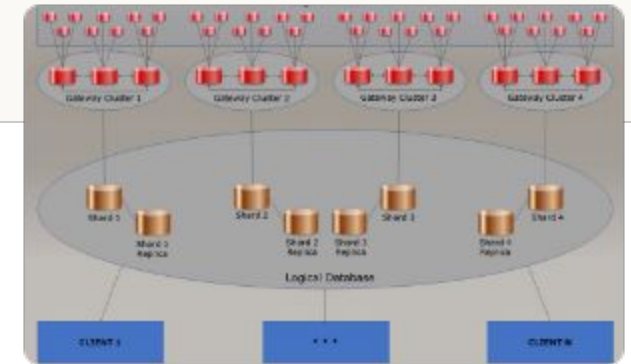
2026-03-06

Dr. IZOUNTAR YOUSRA

What Is a Distributed Database?

General Definition

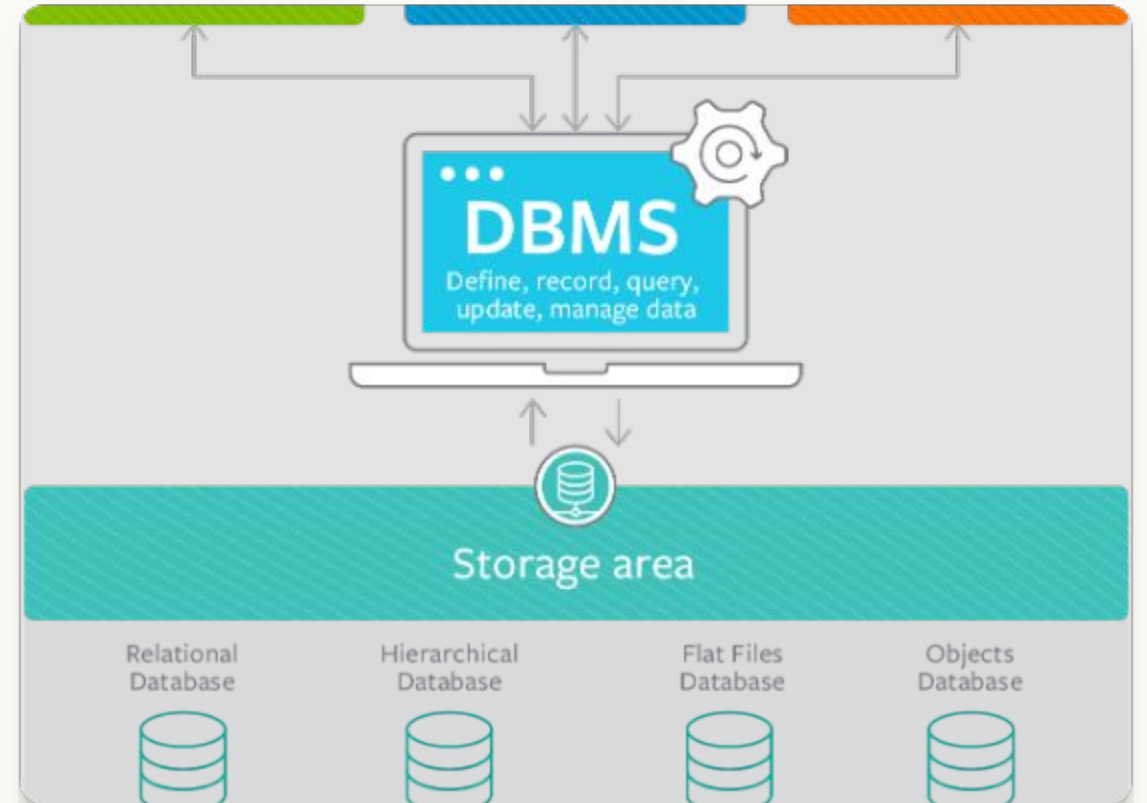
A collection of multiple, logically interrelated databases spread across a computer network.



Database Management System (DBMS / SGBD)

Definition

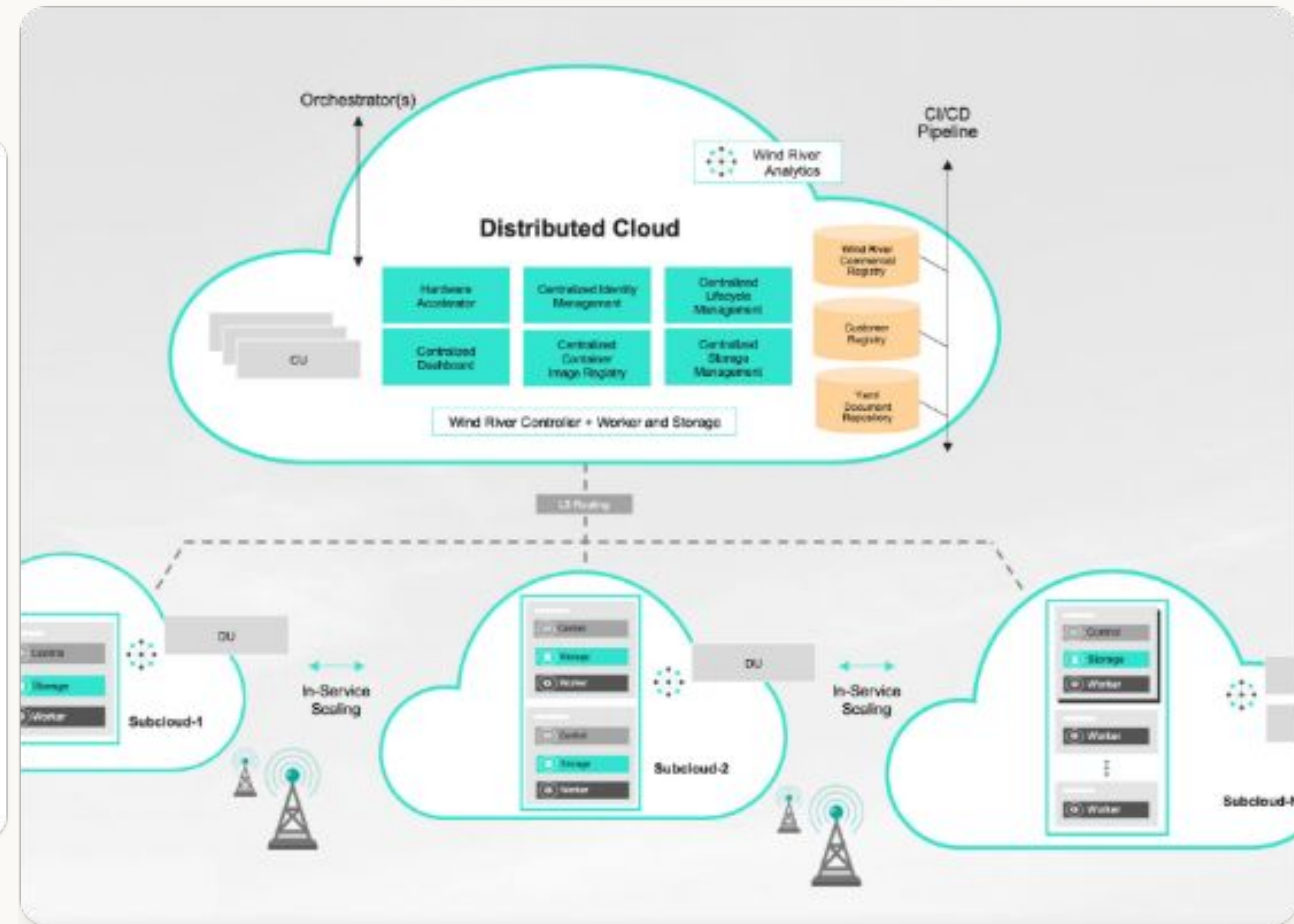
A set of software tools used to manipulate, consult, modify, organize, and protect databases.



Distributed DBMS (DDBMS / SGBDR)

Definition

A system that manages logically linked collections of databases across a network, providing an access mechanism that makes the distribution transparent to users.



Transparency

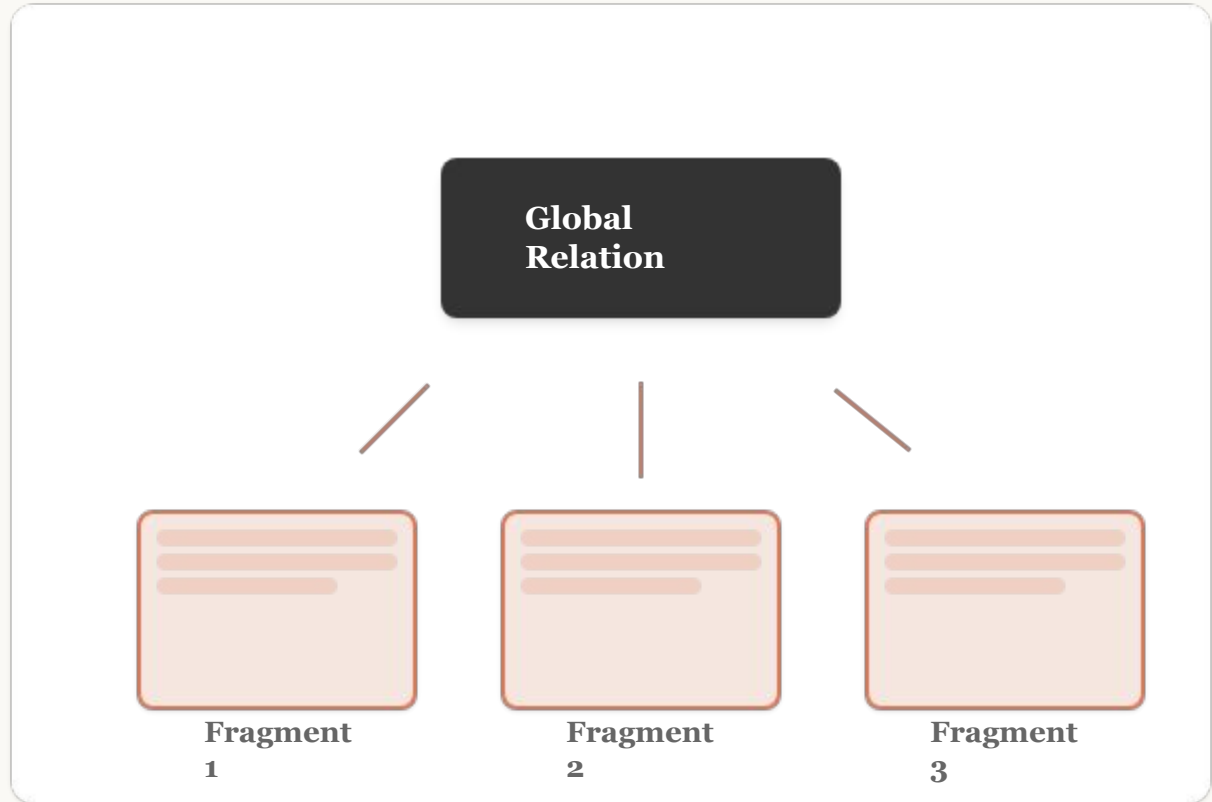
"A core objective where the user perceives the distributed system as one single database without needing to know the physical location of the data."



Fragmentation

Definition

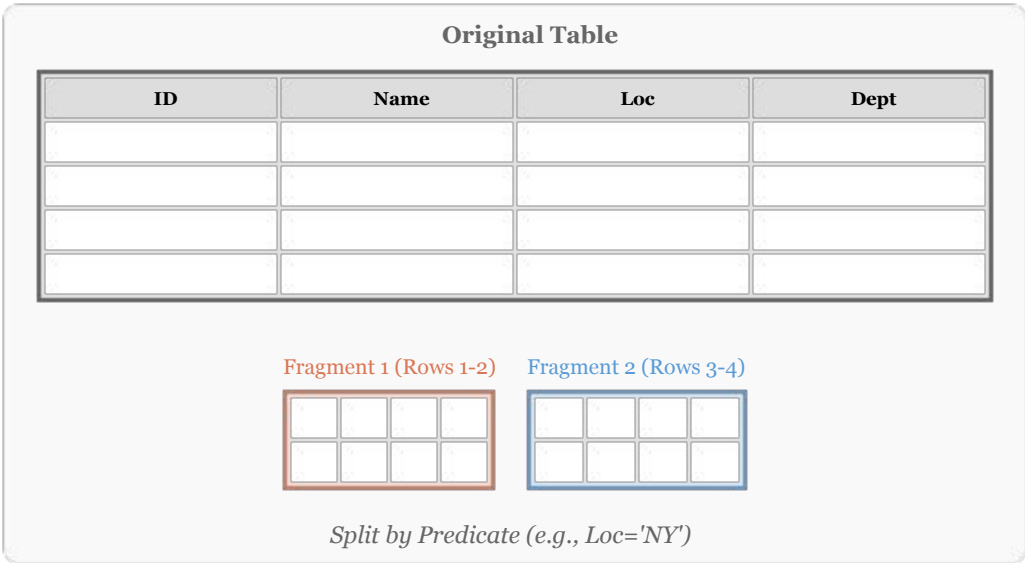
The process of dividing a global relation (table) into smaller pieces called fragments or partitions.



Horizontal vs Vertical Fragmentation

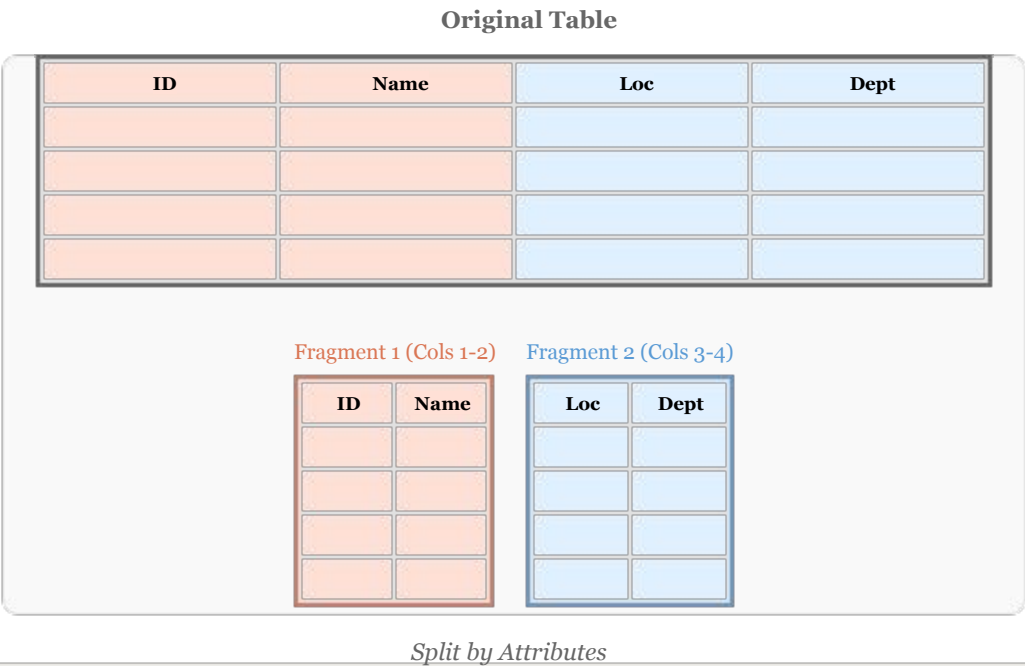
Horizontal Fragmentation

Dividing a table into a set of rows (instances) based on specific criteria or predicates.



Vertical Fragmentation

Dividing a table into a set of columns (attributes).



Mixed & Derived Horizontal Fragmentation

Mixed Fragmentation

A combination where a table is fragmented both horizontally and vertically.

Original
Relation

ID	Name	Loc	Sal



Subset of rows + Subset of columns

Derived Horizontal Fragmentation (DHF)

Fragmenting one table based on the fragmentation scheme of another related table, typically using a foreign key link.

Owner (Parent)

ID	Name	Loc

Fragmented by
Loc='NY'

Link
(FK) →

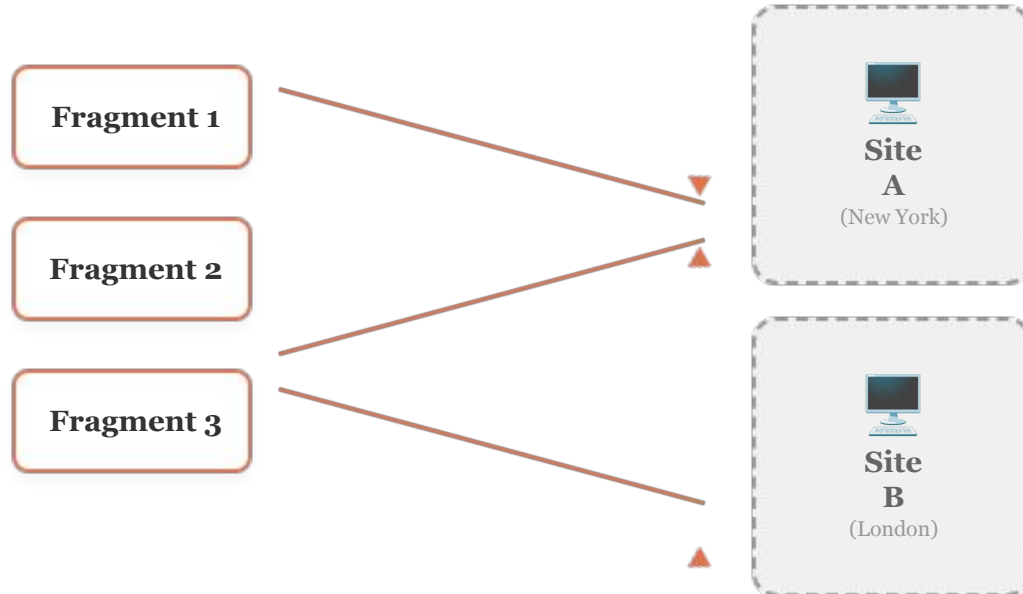
Member
(Child)

ID	OwnID	Date

Fragmented by
Owner.Loc='NY'

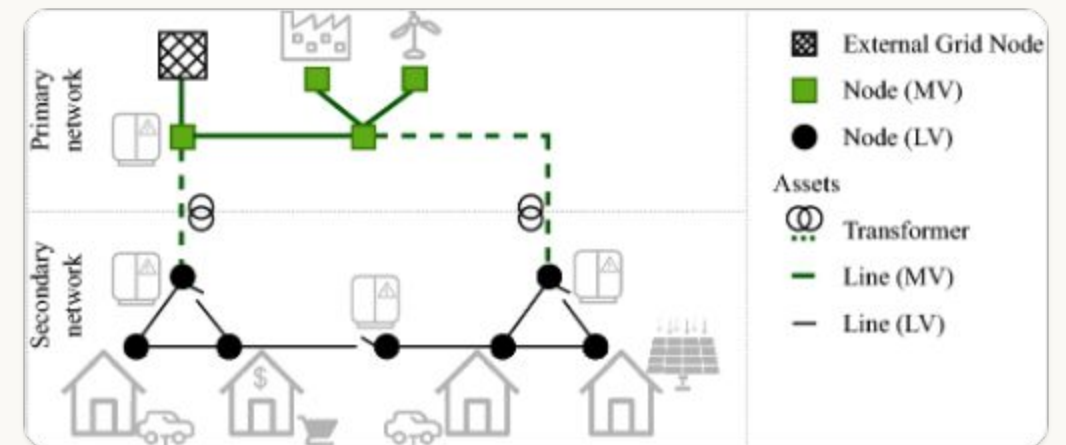
Allocation

Fragment Allocation Map



Definition

- The strategy of deciding which network sites will store specific data fragments.



Replication

Replication Strategy



Fragment A



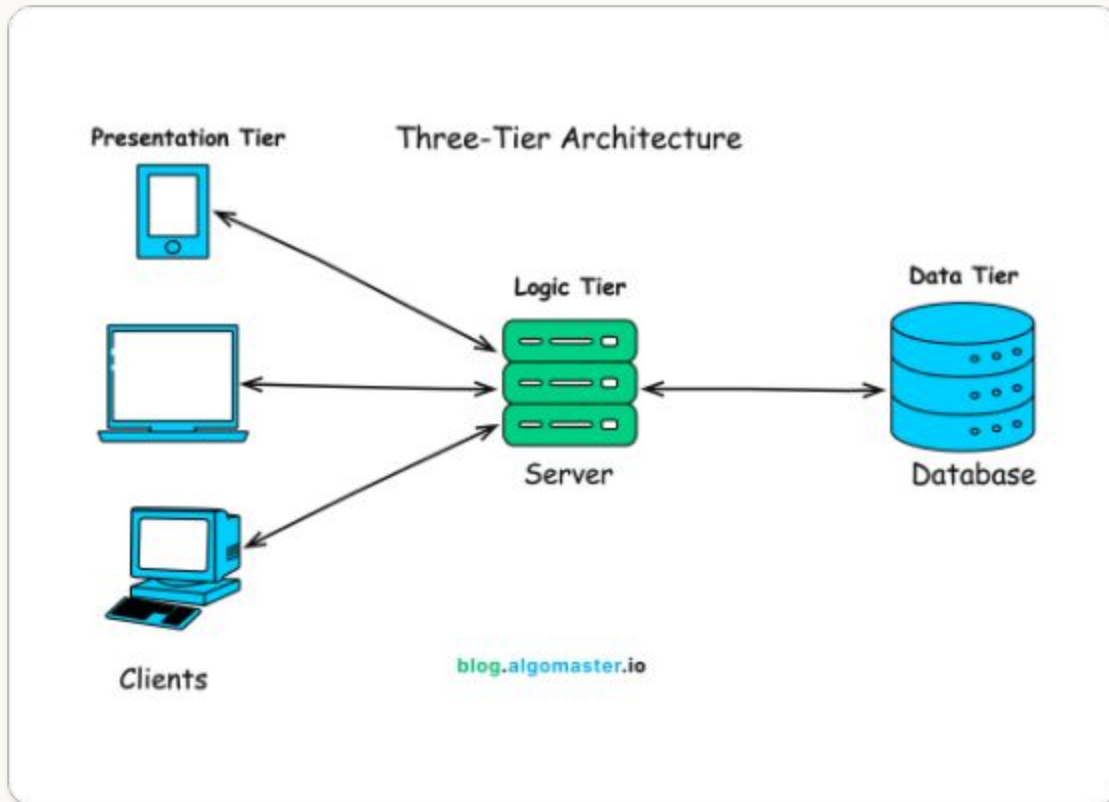
Data remains available via Site 1 & 2 despite Site 3 failure

Core Concept

Creating and maintaining copies of data on multiple sites to ensure data remains available even if one site fails.



Client–Server Architecture



Client ↔ Server with Network Separation

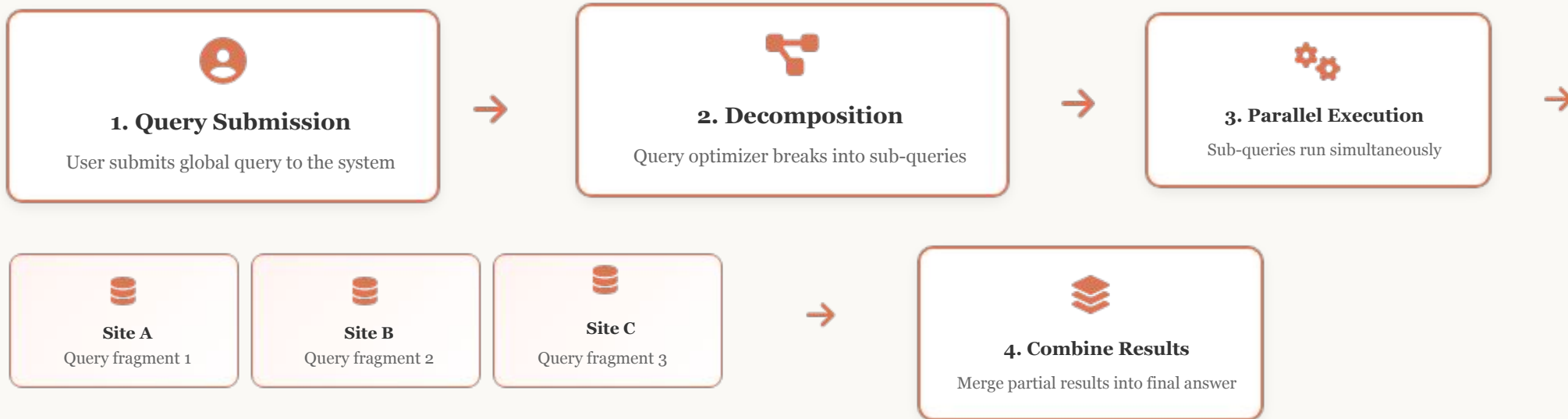
System Definition

- A system where an application program acts as the **client** (handling the user interface and processing).
- The DBMS acts as the **server** (managing data), often residing on different sites.

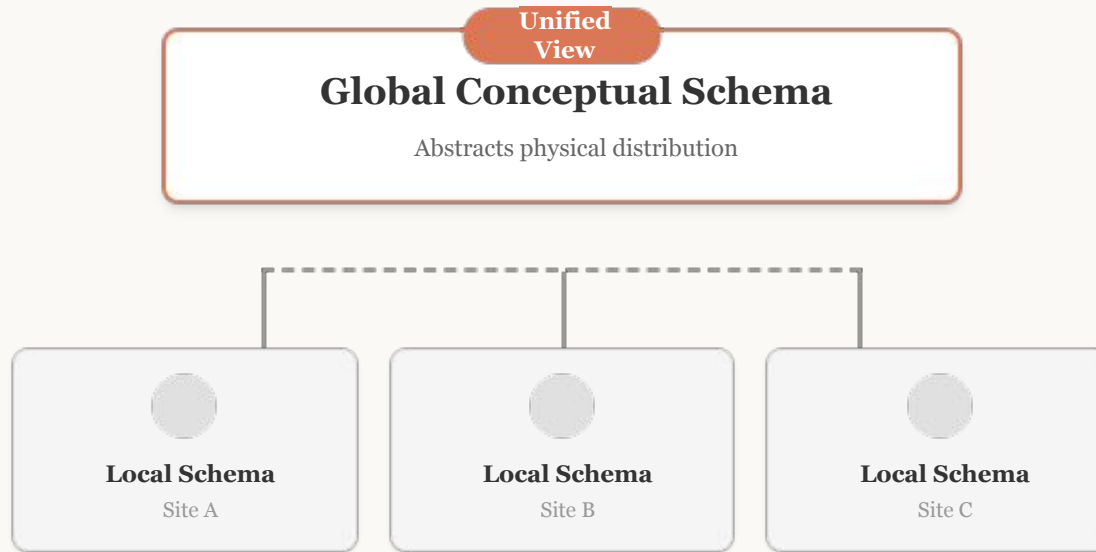
Distributed Query Processing

What is a Distributed Query?

A query that accesses data from multiple sites in a distributed database. The system automatically decomposes it into sub-queries, executes them in parallel across local sites, and combines the results transparently.

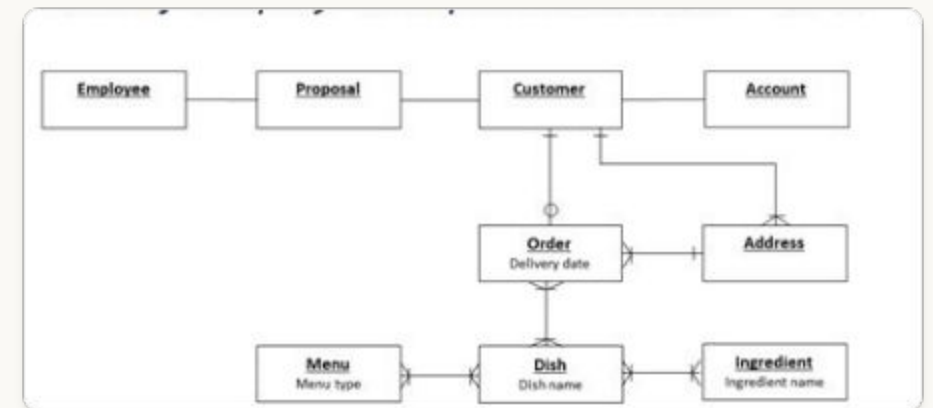


Global Conceptual Schema



Definition

A unified description of the entire distributed database, independent of the physical distribution across sites.



Placement Schema

Fragment Allocation Map

Fragment ID	Relation	Site Location(s)
F1 (North)	Customer	Site 1Site 3
F2 (South)	Customer	Site 2
F3 (East)	Orders	Site 1Site 4
F4 (West)	Orders	Site 3

Example of a Placement Schema Dictionary

Definition

A data dictionary that maps the location of every fragment and replica within the network.

Spreadsheet	Contacts	Corr Address	contacts_tab	cor_address_id	Lookup new address_tab table
Spreadsheet	Contacts	Work Address	contacts_tab	work_address_id	Lookup new address_tab table
Spreadsheet	Contacts	Marital status	contacts_tab	marital_stat	Single, S, Sng => 'S' Married, Marr, M => 'M' Divorced, Div => 'D' Widowed, Wid => 'W'
Spreadsheet	Contacts	Employer	contacts_tab	employer_name	
Spreadsheet	Contacts	Employer	contacts_tab	employer_id	Set if name found in companies_tab
Spreadsheet	Contacts	Job title	contacts_tab	position_text	
Spreadsheet	Contacts	N/A	address_tab	address_id	Generate
Spreadsheet	Contacts	Home, Corr, Work Address	address_tab	address	Extract first line
Spreadsheet	Contacts	Home, Corr, Work Address	address_tab	address_2	Extract second line

Key Concepts Recap



Distributed Database

Multiple interrelated databases across a network, appearing as one.



Distribution Tools

Fragmentation, allocation, replication.



DDBMS Goal

Manage distributed databases with user-facing transparency.



Operation

Client-server use, distributed queries, global + placement schemas.

Catalog

1. Schema & Given Data
2. Queries Q1–Q3: Results
3. Best Horizontal Fragmentation (Customer/Product)
4. Fragments Obtained
5. Query Rewriting (Q1–Q3)
6. Cost Study (Before vs After)
7. Q4: DHF for Join (Customer–Sales)
8. Q5: DHF with Customer & Product + Sales Fragments

Relations and Sample Tuples (Given)

Schema Definitions

- **Customer** (CID, NameC, City)
- **Product** (PID, NameP, Type)
- **Sales** (CID, PID, Qty, UP)

These relations form the basis for the fragmentation exercises. Note the attributes used for filtering in subsequent queries.

Given Tuples: Customer

CID	NameC	City
616	Gilles	Alger
515	Yves	Blida
414	Yamine	Tipaza
313	Ladjel	Tipaza
212	Eric	Alger
111	Pascal	Alger

Given Tuples: Product

PID	NameP	Type
106	Sonoflore	Beauty
105	Clarins	Beauty
104	WebCam	Multimédia
103	Barbie	Toys
102	Engrais	Gardening
101	SlimForm	Fitness

Query Results (Selection Queries)

Q1 Select NameC from Customer where City = 'Alger'

SQL `Select NameC from Customer where City = 'Alger'`

Result Gilles; Eric; Pascal

Q2 Select * from Customer where City = 'Blida'

SQL `Select * from Customer where City = 'Blida'`

Result (515, Yves, Blida)

Q3 Select NameP from Product where Type = 'Beauty'

SQL `Select NameP from Product where Type = 'Beauty'`

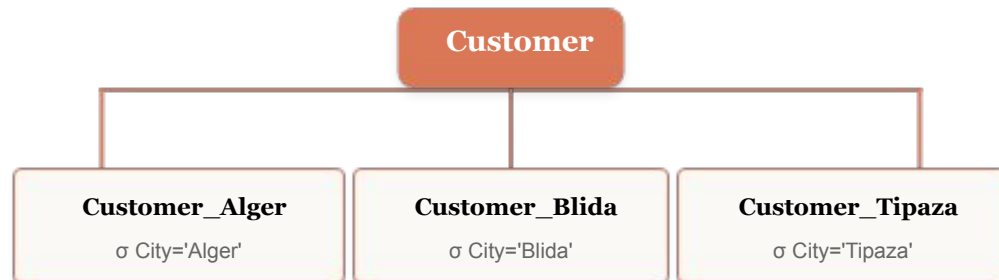
Result Sonoflore; Clarins

Best PHF Choice (Minimize Loaded Tuples)

Goal: Ensure each query loads only the necessary tuples.

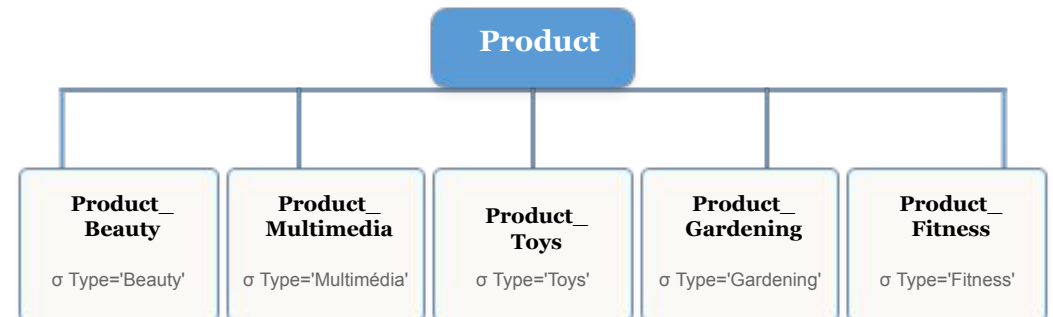
Customer

Primary Horizontal Fragmentation (PHF) by City
(because queries filter by City)



Product

Primary Horizontal Fragmentation (PHF) by Type
(because Q3 filters by Type)



Resulting Fragments for Customer and Product

Customer Fragments

Customer_Alger

CID	NameC	City
616	Gilles	Alger
212	Eric	Alger
111	Pascal	Alger

Customer_Blida

CID	NameC	City
515	Yves	Blida

Customer_Tipaza

CID	NameC	City
414	Yamine	Tipaza
313	Ladjel	Tipaza

Product Fragments

Product_Beauty

PID	NameP	Type
106	Sonoflore	Beauty
105	Clarins	Beauty

Product_Multimedia

PID	NameP	Type
104	WebCam	Multimédia

Product_Gardening

PID	NameP	Type
102	Engrais	Gardening

Product_Toys

PID	NameP	Type
103	Barbie	Toys

Product_Fitness

PID	NameP	Type
101	SlimForm	Fitness

Query Rewriting After PHF

Principle: Rewrite global query to access only the relevant fragment.

Q1 Rewrite

Original

*Select NameC
From Customer...*



Rewritten

**Select NameC
From Customer_Alger**

Q2 Rewrite

Original

*Select *
From Customer...*



Rewritten

**Select *
From Customer_Blida**

Q3 Rewrite

Original

*Select NameP
From Product...*



Rewritten

**Select NameP
From Product_Beauty**

Cost Study - Before vs After

Given Distribution

Algiers:	800,000
Tipaza:	190,000
Blida:	10,000

Total Tuples: 1,000,000

Storage: 1 page = 10 tuples

Total Pages (Before): $1,000,000 / 10 = 100,000$ pages

Q1

Before: 100,000 pages

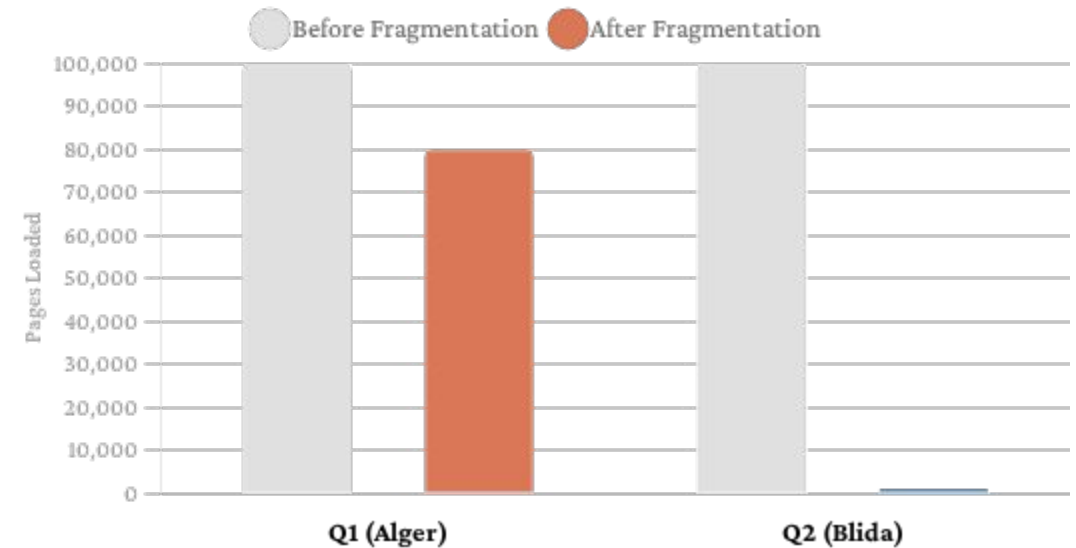
After (Customer_Alger): $800,000 / 10 = 80,000$ pages

Q2

Before: 100,000 pages

After (Customer_Blida): $10,000 / 10 = 1,000$ pages

Execution Cost (Pages Loaded)



Conclusion: Fragmentation reduces useless access; gains are largest for small fragments (e.g., Blida).

Derived Horizontal Fragmentation for a Join (Q4)

Q4 Query

```
Select count(*)  
from Customer c, Sales s  
where s.CID = c.CID and c.City = 'Blida'
```

Best Choice Strategy

Keep **Customer** fragmented by city

Apply **Derived Horizontal Fragmentation (DHF)** to Sales based on Customer fragmentation

Fragmentation Process & Algebraic Expressions



Q4 Execution on Fragments

Obtained Fragments

Customer_Blida		
515	Yves	Blida

Sales Fragment

Sales_Blida (CID=515)			
515	104	5	10
515	102	2	100
515	103	4	17
515	104	7	20

Rewritten Q4

```
Select count (*)
From Sales_Blida s
```

Final Result

count(*) = 4

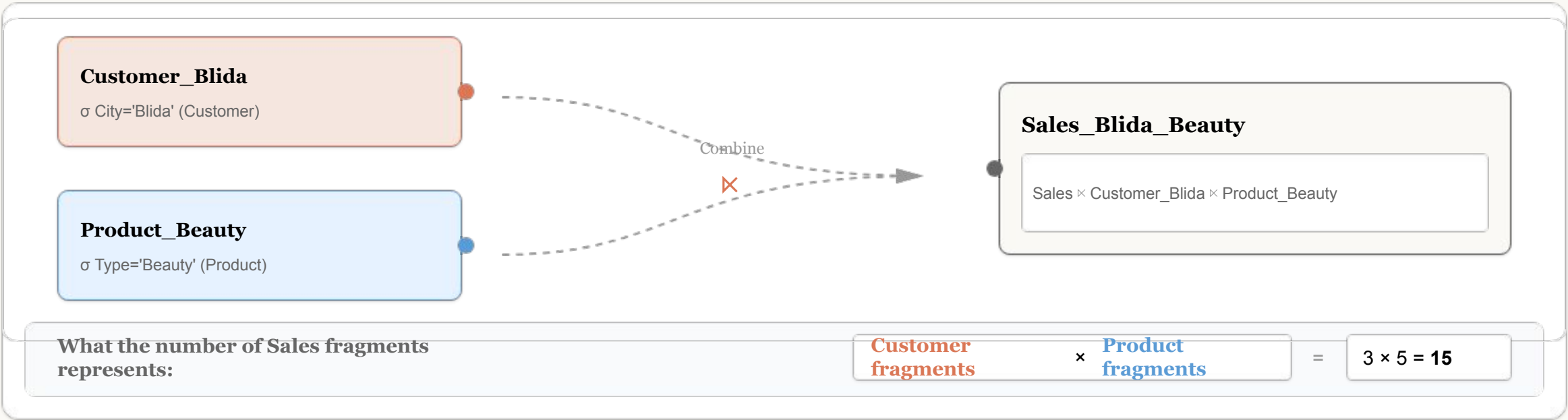
DHF with Customer & Product (Q5)

Q5 Query

```
Select count(*)  
from Customer c, Product p, Sales s  
where s.CID=c.CID and s.PID=p.PID  
and c.City='Blida' and p.Type='Beauty'
```

Best Fragmentation Strategy

Customer fragmented by city
Product fragmented by type
Sales derived fragment for both conditions



Q5 Result Using Derived Sales Fragment

Customer_Blida				Product_Beauty			
515	Yves	Blida		106	Sonoflore	Beauty	
				105	Clarins	Beauty	
Sales rows for customer 515							
515	104	5	10				
515	102	2	100				
515	103	4	17				
515	104	7	20				

Check Condition

Is $PID \in \{105, 106\}$?

$104 \notin \{105, 106\}$

$102 \notin \{105, 106\}$

$103 \notin \{105, 106\}$

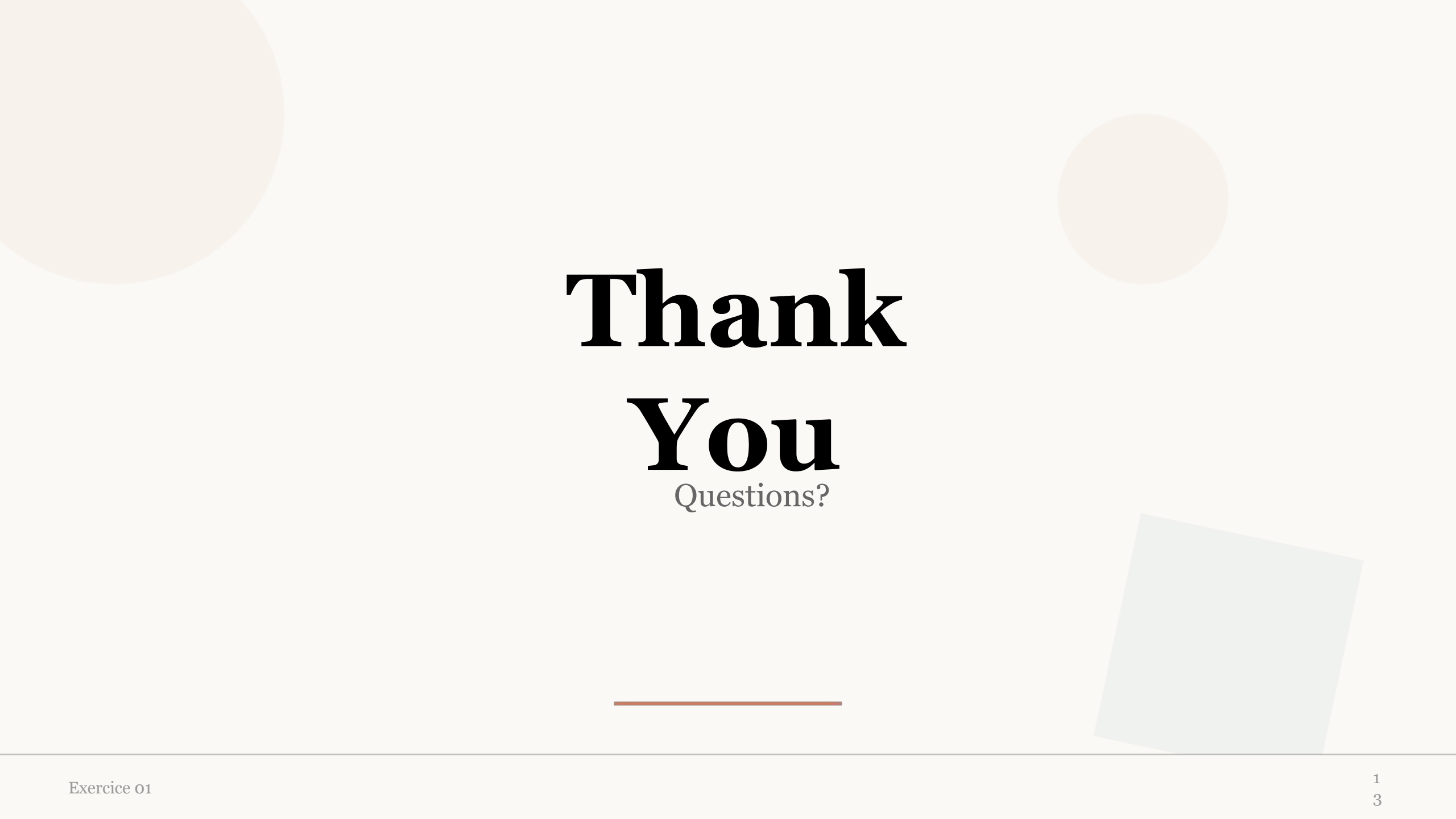
Sales_Blida_Beauty = $\emptyset$

Rewritten Q5

```
Select count (*)
From Sales_Blida_Beauty s,
```

Final Result

count(*) = 0



Thank You

Questions?
